

Requirement Analysis Document (RAD)

Project Title: Digital Shopping Portal (Implementation: "ShopSphere")

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1. Executive Project Summary

ShopSphere is a web-based online shopping system that enables customers to purchase products from different categories through a secure and user-friendly platform. Customers can register, browse products, add items to their shopping cart, place orders, complete online payments, and monitor delivery status. The system also helps administrators manage products, inventory, customer records, and sales information efficiently.

The application uses a **Relational Database Management System (RDBMS)** such as MySQL or PostgreSQL to store customer details, product information, orders, payments, and shipment records. The database ensures accurate data storage and reliable transaction processing.

For example:

- If a customer successfully completes payment, the order is confirmed and stock is updated.
 - If the payment is unsuccessful, the order is cancelled and no changes are made to the inventory.
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2. Core Business Modules

2.1 Customer Management

- New users can register using a unique email address.
- Customers can log in, update their profile, and save delivery addresses.
- Customers can view their previous orders and payment history.

2.2 Product Catalog Management

- Products are organized into categories such as Electronics, Clothing, Grocery, and Home Essentials.

- Customers can search, filter, and view product details.
- The system displays available stock before an order is placed.

2.3 Order & Payment Processing

- Customers can add multiple products to a shopping cart.
- The system supports online payment methods and Cash on Delivery (COD).
- Every successful purchase generates a unique Order ID and invoice.

2.4 Delivery Management

- Delivery details are updated regularly.
 - Customers receive tracking information after the order is shipped.
 - Order status changes from **Placed** → **Confirmed** → **Packed** → **Shipped** → **Delivered**.
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3. System Users (Actors)

Customer

- Register and log in.
- Browse products and place orders.
- Make payments and track deliveries.

Administrator

- Manage products, categories, inventory, and customer accounts.
- Process orders and generate business reports.

Delivery Executive

- View assigned deliveries.
 - Update shipment status.
 - Confirm product delivery.
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4. Reporting Requirements

The system should generate the following reports:

- Monthly Sales Report
- Top Selling Products Report

- Customer Order Report
- Inventory Stock Report

These reports help the business monitor sales performance, manage inventory, and improve customer satisfaction.

5. Non-Functional Requirements

- Customer login information must be secure.
 - Product prices should always be greater than zero.
 - Stock should be updated automatically after each successful order.
 - The system should support multiple users simultaneously.
 - Customer and payment information should remain confidential and protected.
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Alternative Project Titles (if you want something unique):

- **QuickBuy Online Shopping System**
- **BuySmart E-Commerce Platform**
- **NextCart Shopping Portal**
- **MegaMart Online Store**
- **EasyBuy Shopping Management System**
- **ClickCart E-Commerce System**
- **SmartBasket Online Shopping Portal**
- **UrbanCart Shopping System**